14.3.1 Meselson and Stahl's Experiment

Mathew Meselson and Franklin Stahl in 1957 gave experimental evidence for semiconservative replication process.

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Steps

1. *E.coli* cells were grown for many generations in a medium containing radioactive isotope (heavy isotope) of nitrogen source ^{15}N.
2. After many generations all nitrogen containing molecules in *E.coli* cells, including nitrogen bases of DNA contained ^{15}N.
3. The cells labeled with ^{15}N were then transferred to a medium containing only ^{14}N (light isotope). Hence all subsequent DNA synthesised during replication contained ^{14}N.
4. Cell samples were removed at periodic time intervals from the growth medium.
5. From each of the above samples, DNA was isolated and subjected to density gradient centrifugation (Caesium chloride (CsCl) centrifugation).

Expected results

- The heavy isotope ^{15}N containing DNA, will reach equilibrium in a gradient point closer to the bottom of the tube.
- ^{14}N containing DNA will reach equilibrium at a gradient point closer to the surface of the tube.

Observed Results

- After first generation the isolated DNA occupied an intermediate density band.
- After second generation two bands were observed, one at intermediate density and the other at lighter density corresponding to the ^{14}N position in the gradient.

These experimental results and other experiments repeated by Meselson and Stahl with prokaryotes suggested that semiconservative mechanism of replication is universal (Figure 14.12).

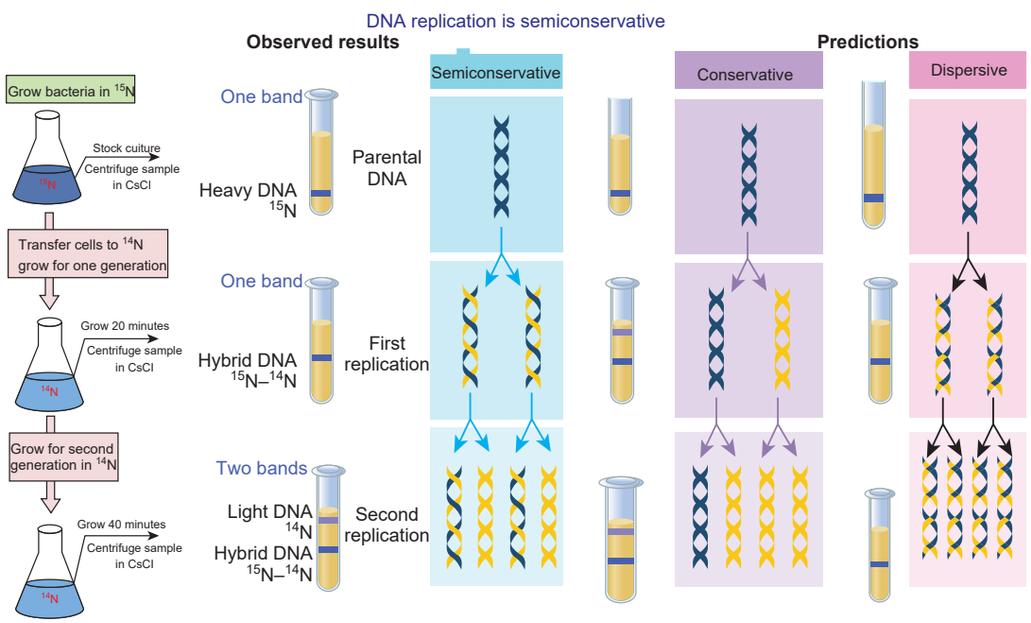


Figure 14.12: Meselson's and Stahl's experiment

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